

Engineering Portfolio

Josiah Christoff Susanto

Electromechanical Systems • CAD Design • Embedded Electronics • Product Development

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Languages:

- Indonesian (Native)
- English (Fluent)
- German (B2)
- Mandarin (Basic)

Featured Project

10m Electronic Target System & 10m Running Target Machine

2023 – Present



Profile

Electrical and Mechanical Engineer with practical experience in electromechanical systems, CAD design, embedded electronics, and product development.

My interest in engineering began through self-directed electronics and mechanical projects. Over time, I developed these skills into commercially successful products, including three 10m Running Target Machine and a 10m Electronic Target System for shooting sports.

I enjoy solving technical problems through prototyping, testing, and iterative design improvements.

Core Skills

Mechanical

- FreeCAD
- 3D Modelling
- Technical Drawings
- Mechanical structure design

Electronics

- STM32
- ESP32
- Arduino
- Soldering

Programming

- Embedded C
- Basic C#
- Basic Java

Project Development

- System Architecture
- Product Development
- Technical Documentation
- Budget Planning

PROJECT OVERVIEW

10m Electronic Target System

Problem

Traditional target systems are expensive and difficult to obtain locally.

Objective

Develop a commercially viable electronic target system capable of supporting shooting sport competitions with low cost and make it affordable for local purchase

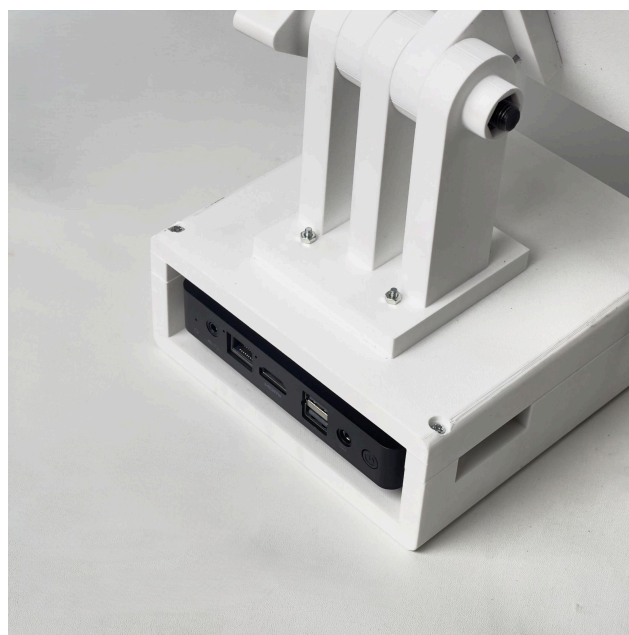
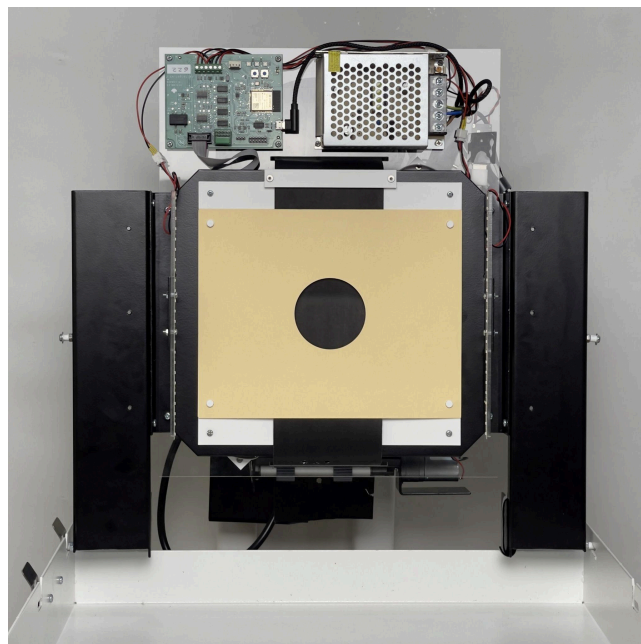
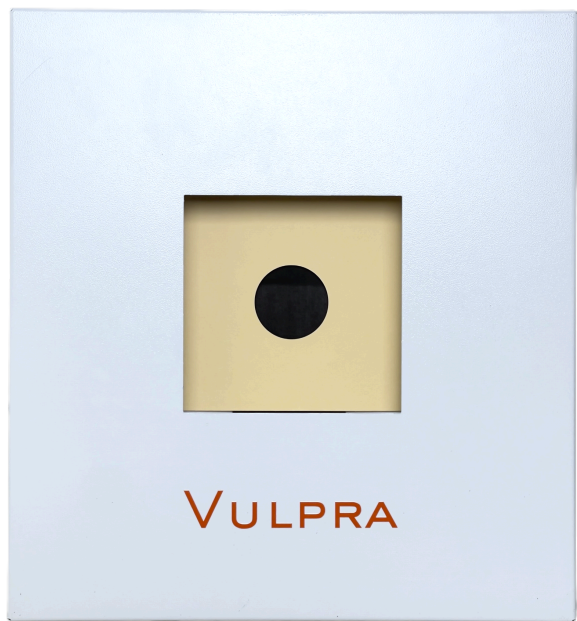
Project Scope

- Mechanical Design
- Electronics Integration
- Embedded Systems
- Assembly Process
- Manufacturing Preparation
- Product Standardization

My Role

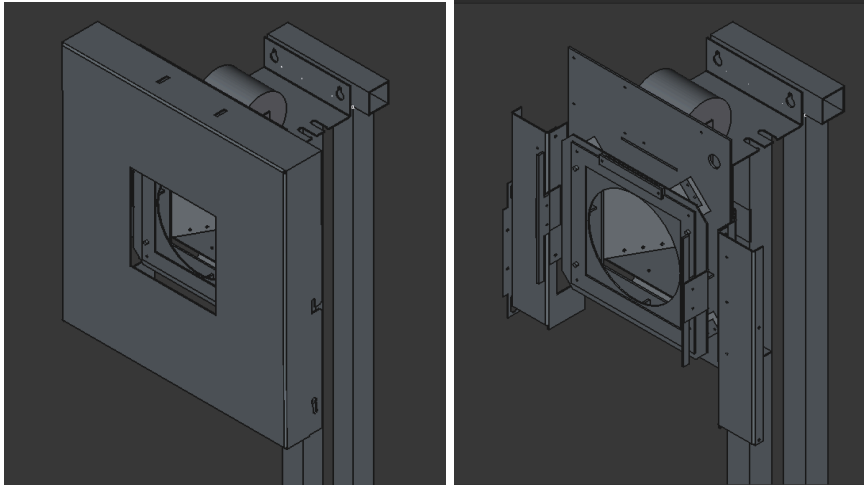
Responsibilities:

- System architecture
- Mechanical design
- Electronics integration
- Logic design
- Assembly
- Testing
- Documentation
- Budget planning
- Customer communication

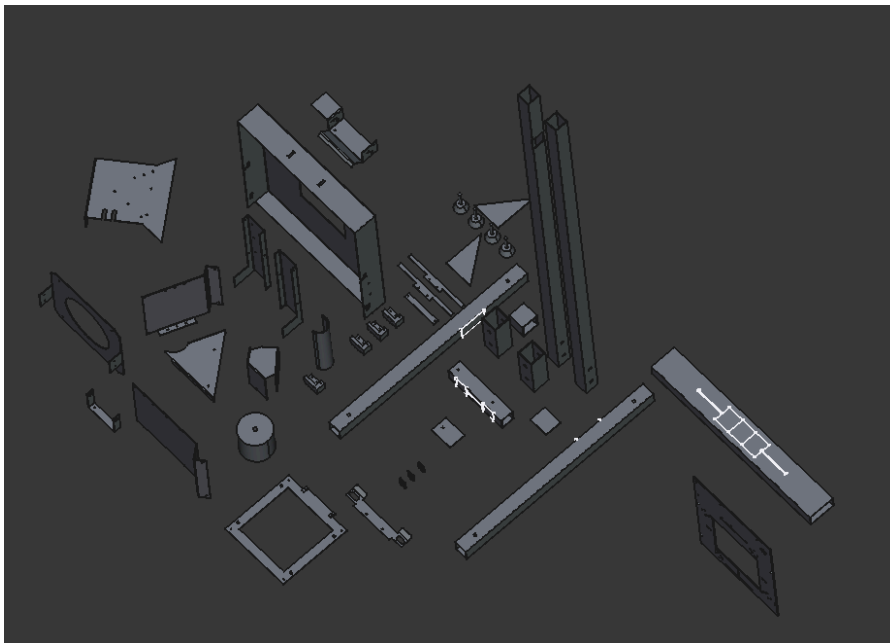


CAD DESIGN

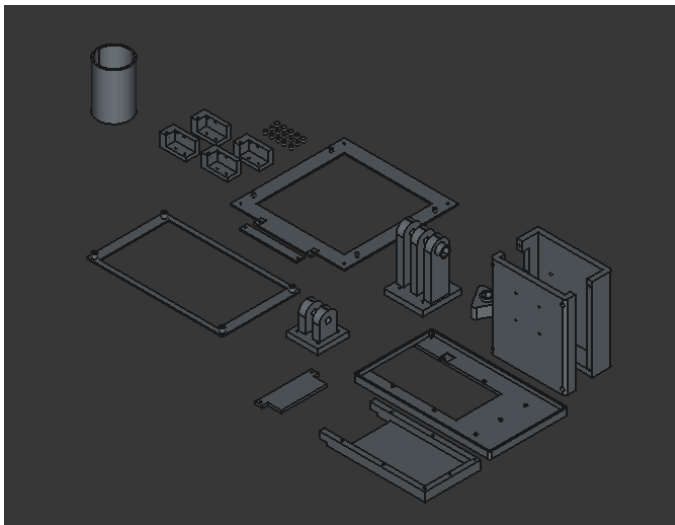
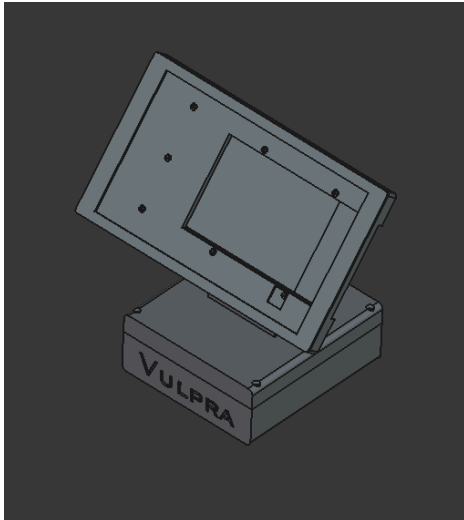
Mechanical Design



This is an overview of the whole mechanical design



This is roughly all the part that i designed



This is the rough overview of the monitor design

Design Considerations

- Ease of manufacturing
- Cost optimization
- Reliability
- Ease of maintenance

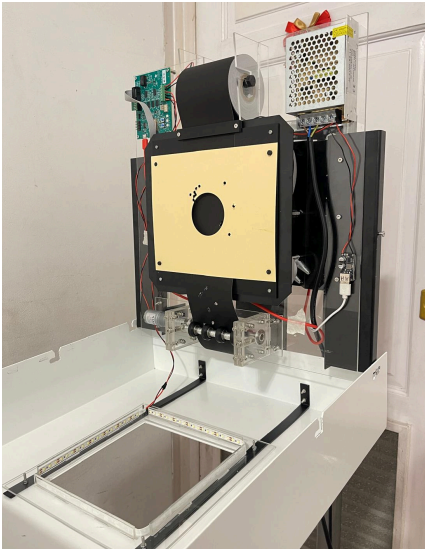
Development Process

Challenge

Create a reliable target system and make it design efficient

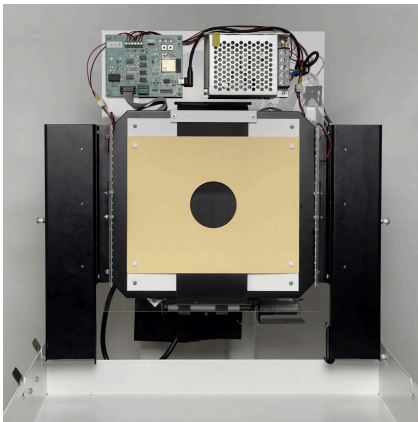
Development

Version 1



Problem : The design is not efficient enough and too complicated. That's why the machine weight and the amount of parts is higher. It is also harder to assemble.

Version 2 (Final Product)



With some improvement and design efficiency, I manage to reduce the parts count, the weight of the machine, and the simplicity of the assembly without reducing the overall function.

This is the overview of the internal component before fully assembled.

10m Running Target

The project documentation is in this google drive link :

📁 Running target portfolio

Thank You

For further information, CAD files and technical documentation:

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Portfolio Repository: https://github.com/redfoxxxi/ET_MASTER_CODE